

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : Cyber & Information Security

AIM : Configure and test firewall rules to control network traffic, filter packets based on specified criteria, and protect network resources from unauthorized access.

Steps:

1. Open Windows Firewall

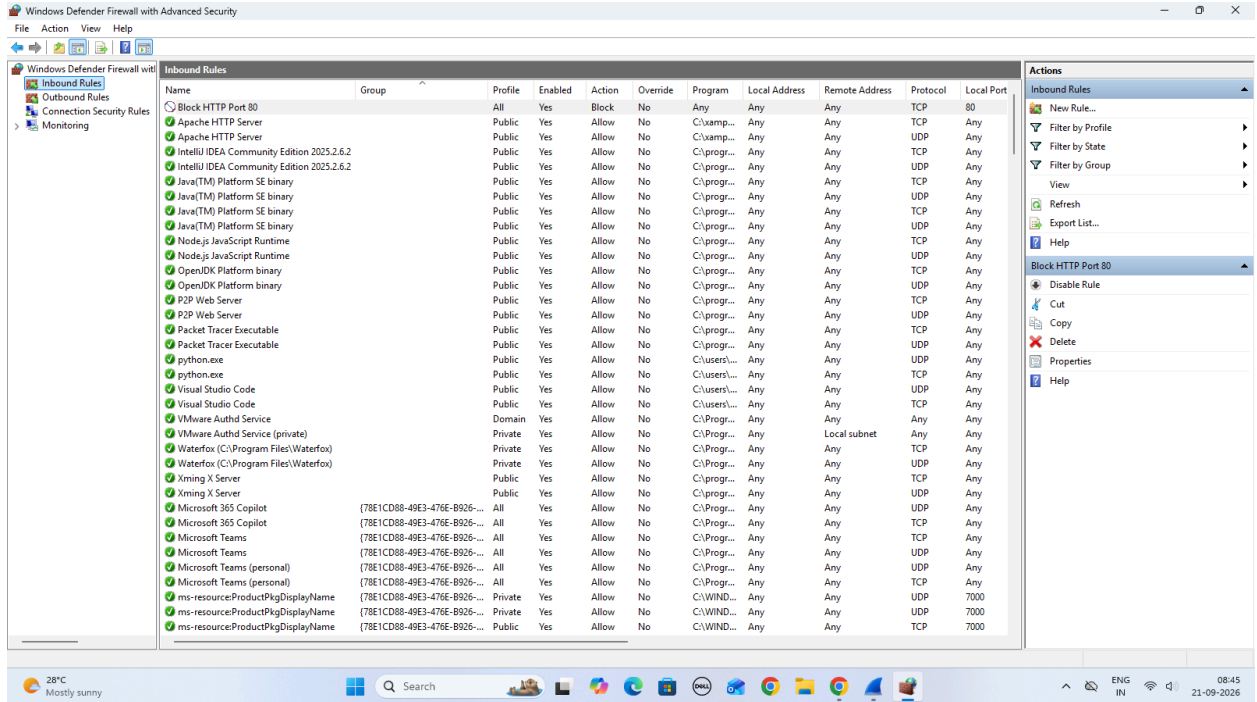
1. Press Windows + R.
2. Type: wf.msc
3. Press Enter.
4. Windows Defender Firewall with Advanced Security will open.

2. Create an Inbound Firewall Rule

1. Click **Inbound Rules** on the left.
2. Click **New Rule...** on the right.
3. Select **Port** → **Next**.
4. Select **TCP**.
5. Enter a port, for example: 80
6. Click **Next**.
7. Select **Block the connection**.
8. Click **Next**.
9. Select **Domain, Private, Public**.
10. Give the rule a name: Block HTTP Port 80

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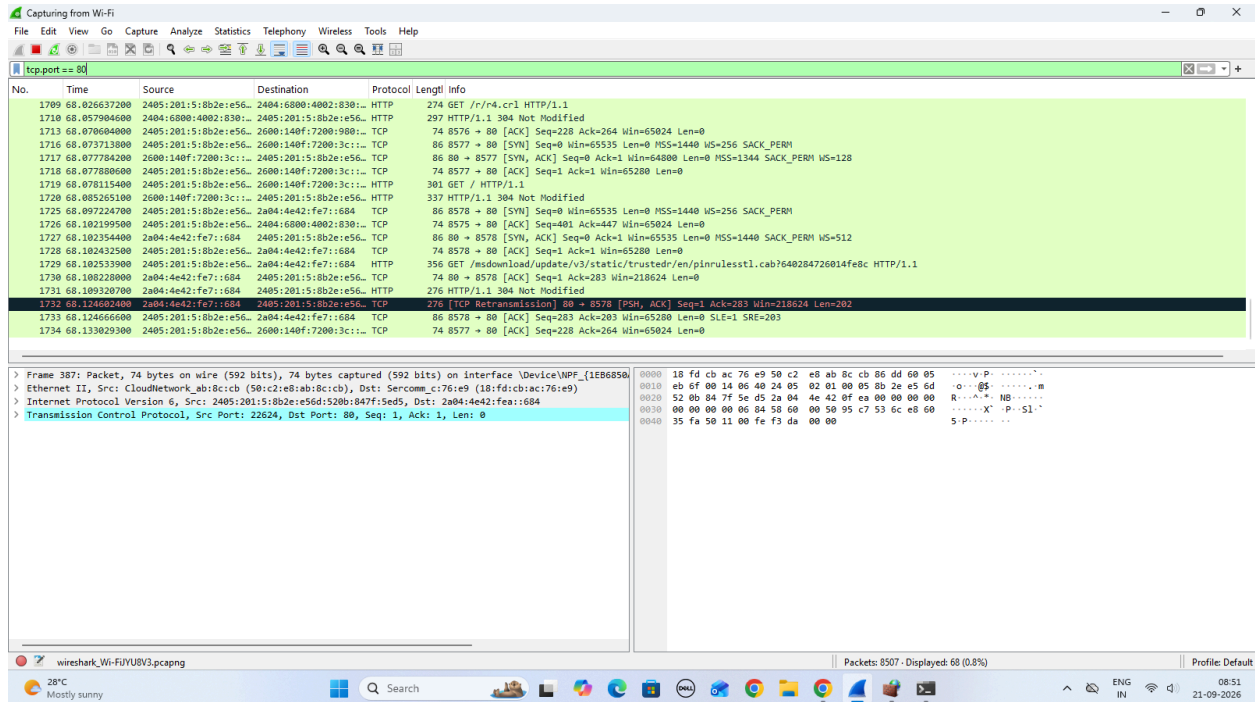
11.

3. Start Wireshark

1. Open **Wireshark**.
2. Select your active network interface:
 - o Wi-Fi, or
 - o Ethernet
3. Click **Start Capturing Packets**.

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4.Filter Packets in Wireshark

To view HTTP traffic: **tcp.port == 80**

To view TCP traffic: **tcp**

To view traffic from a particular IP: **ip.addr == 192.168.1.10**

To view blocked-port traffic: **tcp.port == 80**

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The image shows a Wireshark packet capture window titled "Capturing from Wi-Fi". The filter is set to "tcp.port == 80". The packet list shows several packets, with packet 387 selected. The packet details pane shows the following information:

- Frame 387: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{1E68580...}
- Ethernet II, Src: CloudNetwork_ab8c:cb (50:c2:e8:ab:8c:cb), Dst: Sercomm_c:76:e9 (18:fd:cb:ac:76:e9)
- Internet Protocol Version 6, Src: 2405:201:5:8b2e:e56d:520b:847f:5ed5, Dst: 2a04:4e42:fea1:684
- Transmission Control Protocol, Src Port: 22624, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, IP header, and TCP header.

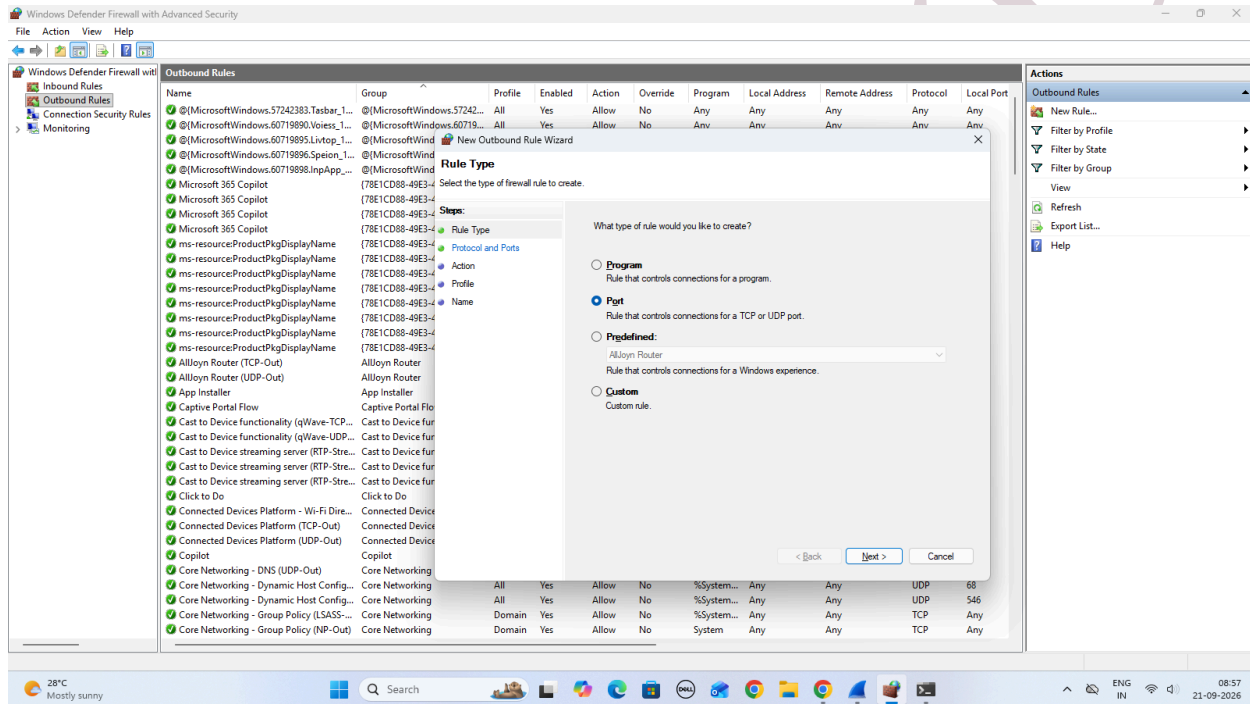
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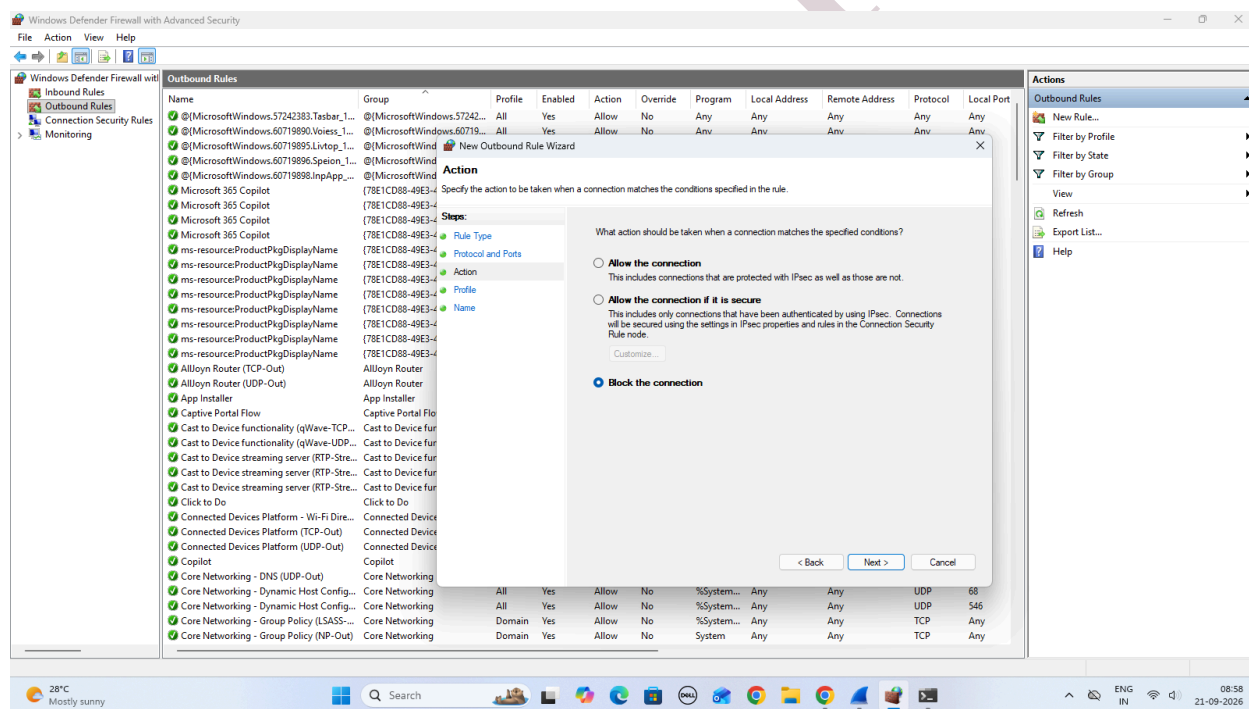
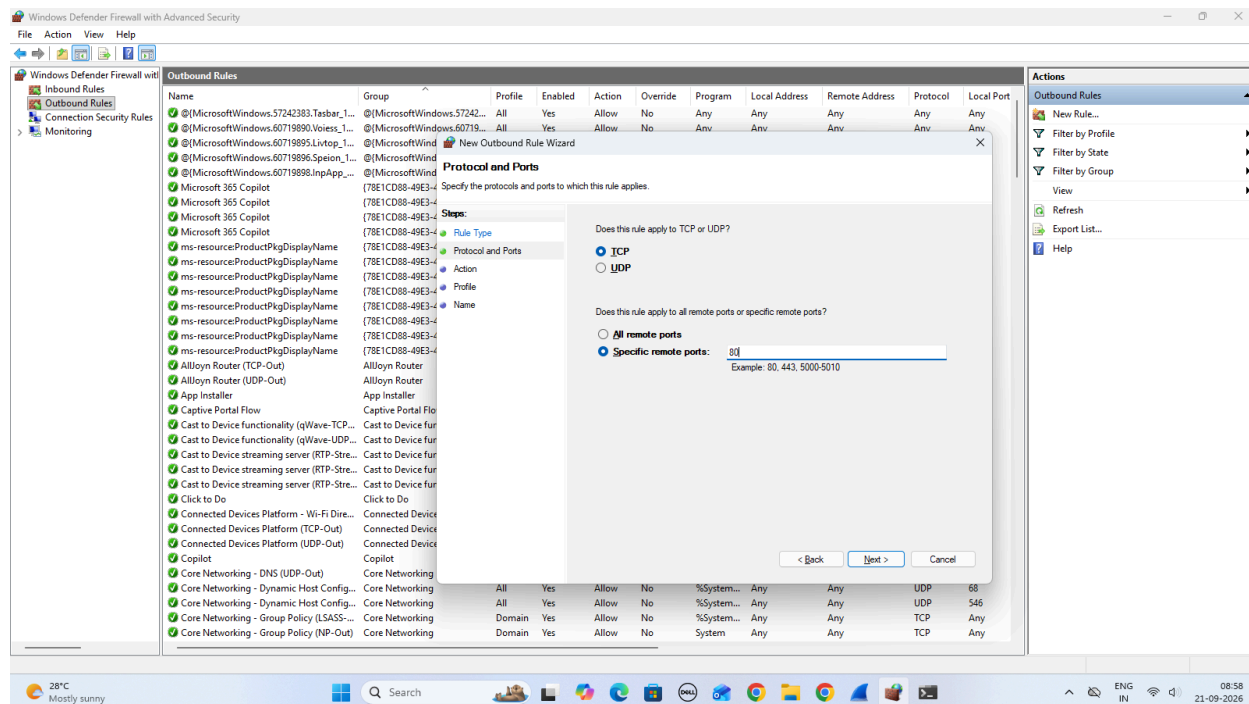
6. Create an Outbound Rule

1. Open **Windows Defender Firewall with Advanced Security**.
2. Select **Outbound Rules**.
3. Click **New Rule...**
4. Select **Port**.
5. Select **TCP**.
6. Enter: **80**
7. Select **Block the connection**.
8. Select all profiles.
9. Name it: **Block Outbound HTTP**



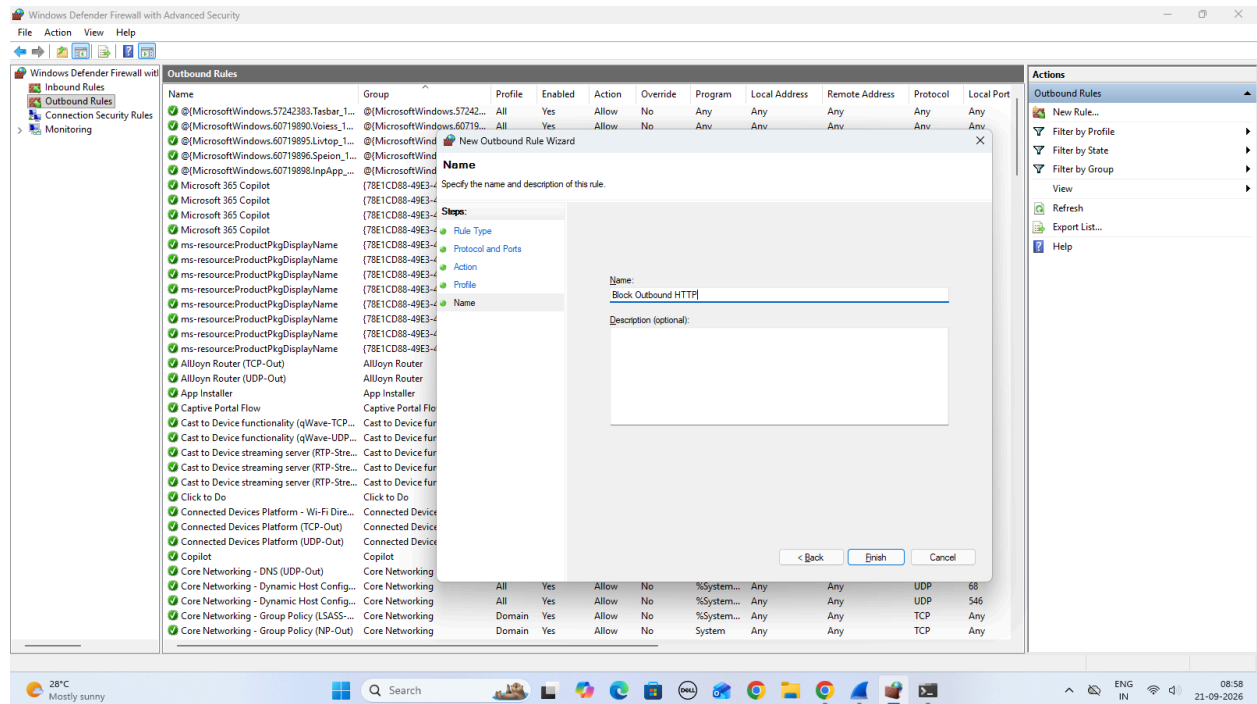
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7. Observe in Wireshark

In Wireshark, use: **tcp.port == 80**

You can observe TCP packets such as:

1. SYN
2. SYN, ACK
3. ACK

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tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
387	9.864892300	2405:2011:5:8b2e:e56...	2a04:4e42:fea1::684	TCP	74	22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
388	9.865063600	2405:2011:5:8b2e:e56...	2a04:4e42:fea1::684	TCP	74	22626 → 80 [FIN, ACK] Seq=1 Ack=1 Win=255 Len=0
392	9.873725200	2a04:4e42:fea1::684	2405:2011:5:8b2e:e56...	TCP	74	80 → 22626 [ACK] Seq=1 Ack=2 Win=427 Len=0
393	9.873725600	2a04:4e42:fea1::684	2405:2011:5:8b2e:e56...	TCP	74	80 → 22626 [FIN, ACK] Seq=1 Ack=2 Win=427 Len=0
394	9.873811400	2405:2011:5:8b2e:e56...	2a04:4e42:fea1::684	TCP	74	22626 → 80 [ACK] Seq=2 Ack=2 Win=255 Len=0
395	10.767008800	2405:2011:5:8b2e:e56...	2a04:4e42:fea1::684	TCP	74	[TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
410	10.767854400	2405:2011:5:8b2e:e56...	2a04:4e42:fea1::684	TCP	74	[TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
415	11.993424700	2405:2011:5:8b2e:e56...	2a04:4e42:fea1::684	TCP	74	[TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
630	14.397960900	2405:2011:5:8b2e:e56...	2a04:4e42:fea1::684	TCP	74	[TCP Retransmission] 22624 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
733	17.581346900	2405:2011:5:8b2e:e56...	2606:4700:b7d5:72db...	TCP	86	8573 → 80 [SYN] Seq=0 Min=65535 Len=0 MSS=1440 WS=256 SACK_PERM
734	17.590135500	2606:4700:b7d5:72db...	2405:2011:5:8b2e:e56...	TCP	86	80 → 8573 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1360 SACK_PERM WS=6192
735	17.598233000	2405:2011:5:8b2e:e56...	2606:4700:b7d5:72db...	TCP	74	8573 → 80 [ACK] Seq=1 Ack=1 Win=65200 Len=0
736	17.598044100	2405:2011:5:8b2e:e56...	2606:4700:b7d5:72db...	HTTP	149	GET / HTTP/1.1
737	17.597886800	2606:4700:b7d5:72db...	2405:2011:5:8b2e:e56...	TCP	74	80 → 8573 [ACK] Seq=1 Ack=76 Win=131072 Len=0
738	17.601608800	2606:4700:b7d5:72db...	2405:2011:5:8b2e:e56...	TCP	943	80 → 8573 [PSH, ACK] Seq=1 Ack=76 Win=131072 Len=869 [TCP PDU reassembled in 739]
739	17.601681200	2606:4700:b7d5:72db...	2405:2011:5:8b2e:e56...	HTTP	79	HTTP/1.1 200 OK (text/html)
740	17.601675800	2405:2011:5:8b2e:e56...	2606:4700:b7d5:72db...	TCP	74	8573 → 80 [ACK] Seq=76 Ack=875 Win=64512 Len=0
741	17.601865400	2405:2011:5:8b2e:e56...	2606:4700:b7d5:72db...	TCP	74	8573 → 80 [FIN, ACK] Seq=76 Ack=875 Win=64512 Len=0
745	17.608989800	2606:4700:b7d5:72db...	2405:2011:5:8b2e:e56...	TCP	74	80 → 8573 [FIN, ACK] Seq=875 Ack=877 Win=131072 Len=0

> Frame 387: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{1EB86580} ...

> Ethernet II, Src: CloudNetwork_ab8c8bcb (50:c2:e8:ab:8c:bcb), Dst: Sercomm_c76:e9 (18:fd:cb:ac:76:e9) ...

> Internet Protocol Version 6, Src: 2405:2011:5:8b2e:e56d:520b:b47f:5ed5, Dst: 2a04:4e42:fea1::684 ...

> Transmission Control Protocol, Src Port: 22624, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

```

0000  18 fd cb ac 76 e9 50 c2 e8 ab 8c bc 86 dd 68 05  ....P.....
0010  eb 6f 00 14 06 04 24 05 02 01 05 8b 2e e5 6d  ....@.....m
0020  52 00 84 7f 5e d5 2a 04 4e 42 0f ea 00 00 00  R.....NB....
0030  00 00 00 00 06 84 50 00 00 59 c7 53 6c e0 60  ....X'P.....
0040  35 fa 50 11 00 fe f3 da 00 00  ....S-P.....

```

wireshark-Wi-Fi\UBV3.pcapng
Packets: 19138 - Displayed: 94 (0.5%)
Profile: Default